$$\text{Cycle: } (2^S - 3^k)n = C > 0$$

$$\text{Same word, } n > m\text{: } 3^k(n-m) = 2^S(y-x)$$

$$2^S \mid (n-m) \;\Rightarrow\; n - m \geq 2^S$$

If $2^s \geq n$, no smaller positive m can share that word.